

IS4007 - STATISTICS IN PRACTICE II

Customer Segmentation And Revenue Forecasting

A Retail Analytics Study



Executive Summary

This report brings together two linked analytical pieces carried out on the company's transaction data: a customer segmentation study and a revenue forecasting study. Both draw on the same underlying sales records-2,000 customers and roughly 50,000 purchases across 24 months (January 2024 to December 2025)-but they answer different questions. The first half of the report asks who our customers are and how much they are worth; the second half asks what the business's revenue is likely to do over the next six months, and how much that forecast can be trusted.

Part I, Customer Segmentation Analysis, uses Recency-Frequency-Monetary (RFM) scoring and K-means clustering to move beyond the company's existing Premium / Standard / Budget / New labels and group customers by what they actually do. It identifies 504 High Value, 845 Medium Value, and 651 Low Value customers; finds that the existing labels frequently disagree with real spending behavior; confirms four natural customer groups (Champions, Repeat Customers, Occasional, and an At-Risk group of 114 customers); estimates long-term customer value (CLV); and shows that revenue is spread broadly across the customer base rather than concentrated in a handful of big spenders.

Part II, Revenue Forecasting, builds a classical trend-plus-seasonal model on 24 months of total monthly revenue, tests it honestly against three held-out months, and uses it to forecast January–June 2026. The model cuts forecast error from 26.4% (a naive flat-average guess) to 2.1% on unseen data, and is stress-tested with four residual diagnostic checks so that its limitations are stated openly rather than hidden.

Read together, the two studies point in the same direction: the business has a healthy, broad-based customer portfolio and a revenue forecasting model that is accurate enough to plan around, but both come with clearly flagged caveats - a small but valuable at-risk customer segment that needs a win-back effort, and a forecast whose confidence range should be read as approximate rather than statistically exact.

Data Description

The analysis is based on a single transaction-level dataset, *master_sales.csv*, containing 50,000 individual sales transactions recorded across 2,000 unique customers between 1 January 2024 and 31 December 2025 (24 full calendar months). Each row represents one transaction and carries 40 fields, which fall into five broad groups:

Category	Fields	Description
Transaction identifiers & timing	transaction_id, transaction_date, transaction_year, transaction_month, transaction_year_month	Uniquely identifies each sale and its date, with pre-derived year/month breakdowns
Customer information	customer_id, customer_type, segment, acquisition_channel, customer_since, customer_tenure_days, location_category, customer_attribute, attribute_type, active_status	Describes who the customer is - individual/SME/corporate, their existing Premium/Standard/Budget/New label, how and when they were acquired, and their location type
Product information	product_id, product_name, category, standard_price, cost_price, margin_percentage	Identifies the item sold, its product category, and its standard pricing/cost structure
Transaction financials	quantity, unit_price, discount_percentage, discount_band, gross_revenue, net_revenue, cost, gross_margin, margin_percentage_tx, payment_status	The actual pricing, discounting, and profitability of that specific transaction
Sales channel & campaigns	channel, branch_id, campaign_id, has_campaign, campaign_name, campaign_start_date, campaign_end_date, offer_type, campaign_cost	Where the sale took place (online, branch, dealer, direct sales, or partner) and whether it was tied to a promotional campaign

For this study, the customer segmentation module (Part I) was built primarily from customer_id, transaction_date, net_revenue, and customer_since, aggregated to one row per customer. The revenue forecasting module (Part II) was built from transaction_date and net_revenue, aggregated to one total per calendar month. The remaining fields - product, campaign, channel, and pricing detail - were not used in the current analysis but remain available in the dataset for future work (e.g., product-level or campaign-level performance analysis).

Part I

Customer Segmentation Analysis

1. What This Is About

Every time a customer makes a purchase, it is recorded as a transaction. Over time, these records build up a rich picture of how each customer behaves -how often they buy, how much they spend, and whether they are still buying at all.

Rather than relying on gut feel, this analysis uses that transaction history to answer four practical questions:

- Who are our most valuable customers, based on what they actually do -not just how we've labelled them?
- Are there natural "types" of customers that emerge from the data on their own?
- What is each customer likely to be worth to the business over the next few years?
- Which customers have gone quiet, and how much revenue is riding on winning them back?

2. Measuring Customer Value: The "RFM" Method

To compare customers fairly, we measured three simple things for every customer:

- Recency -how many days ago they last made a purchase. A smaller number is better (they bought recently).
- Frequency -how many separate purchases they have made in total.
- Monetary -how much money they have spent with us in total.

Together, these three measures are known as "RFM" -a widely used, straightforward way to describe customer behavior without needing complex models.

2.1 Recency, Frequency, and Monetary (RFM) Overview

Table 1: Typical Customer Values (Recency, Frequency, Monetary)

Measure	Typical customer
Days since last purchase (median)	16 days
Number of purchases made (median)	25 purchases
Total amount spent (median)	LKR 8,781
Highest total spent (single customer)	LKR 23,401

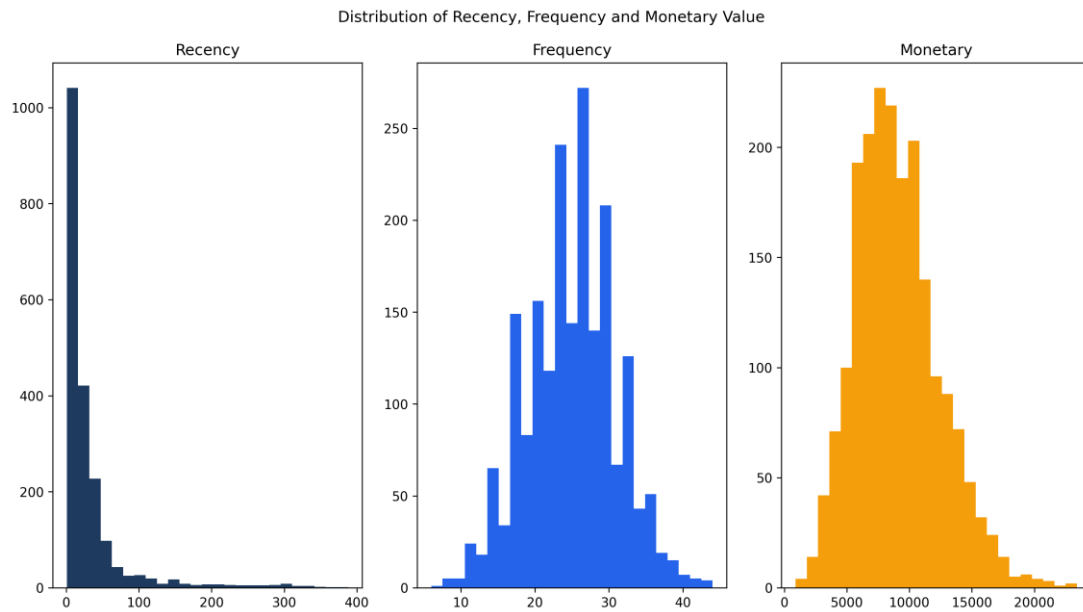


Figure 1: How Recency (days since last purchase), Frequency (number of purchases), and Monetary (total spent) are spread across the 2,000 customers. Most customers cluster on the left of the Recency chart -meaning most people have bought fairly recently -while spending and purchase counts are more evenly spread.

2.2 Turning RFM into a simple score

To make the three measures comparable, each customer was ranked and split into five equal-sized groups for each measure, with scores ranging from 1 (worst) to 5 (best). A customer's three scores are then added together, giving a combined score between 3 (lowest) and 15 (highest). That combined score is used to sort customers into three easy-to-use tiers:

- High Value -combined score of 12 or more (504 customers, 25% of the base)
- Medium Value -combined score of 8 to 11 (845 customers, 42% of the base)
- Low Value -combined score of 3 to 7 (651 customers, 33% of the base)

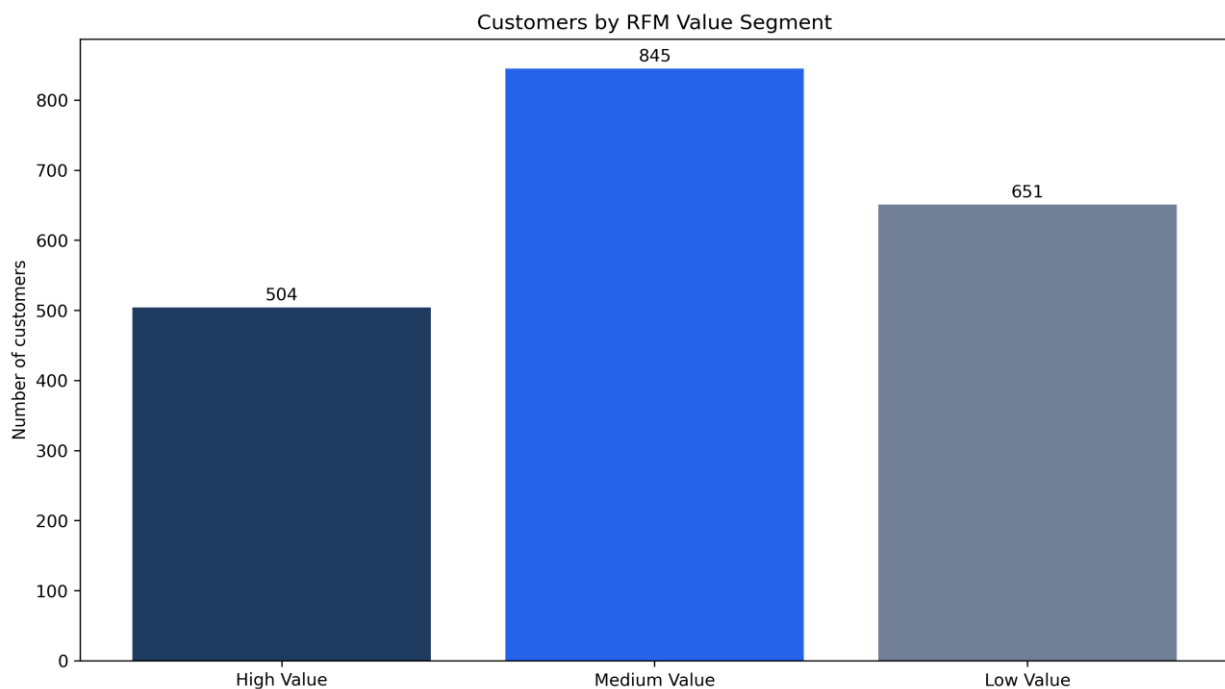


Figure 2: Number of customers in each value tier, based on actual buying behavior.

3. Does Our Existing Labelling Match Reality?

The business already labels every customer as Premium, Standard, Budget, or New. This section checks whether that label actually reflects how much a customer spends and how often they buy -or whether it's just a name that was assigned when they first signed up.

Table 2: RFM Value Tier vs. Business Segment Label

Behavior-based tier	Budget	New	Premium	Standard
High Value	131	75	73	225
Medium Value	207	119	128	391
Low Value	163	98	102	288

The table above compares the two views side by side. If the existing labels perfectly matched behavior, almost all "Premium" customers would sit in the High Value row, and almost all "Budget" customers would sit in the Low Value row. That is not what we see.

- 225 "Standard" customers and 131 "Budget" customers are, in fact, High Value -they are spending like top customers despite their label.
- 102 customers labelled "Premium" fall into the Low Value tier, despite carrying a premium name.

The existing Premium / Standard / Budget label should not be used on its own to decide who gets priority service or targeted offers. A customer's actual buying behavior is a more reliable guide to their real value than the label they were given at sign-up.

4. Letting the Data Find Its Own Customer Groups

Section 2 split customers into three tiers using fixed cut-off scores (12 and 8) that we chose by hand. This section asks a different question: if we let a computer look purely at the numbers -with no pre-set cut-offs -what natural groupings does it find on its own? This technique is called clustering, and the specific method used is called K-means.

In simple terms, K-means looks at where customers sit on the Recency, Frequency, and Monetary measures, and groups together customers who are numerically close to one another. We first needed to decide how many groups to look for.

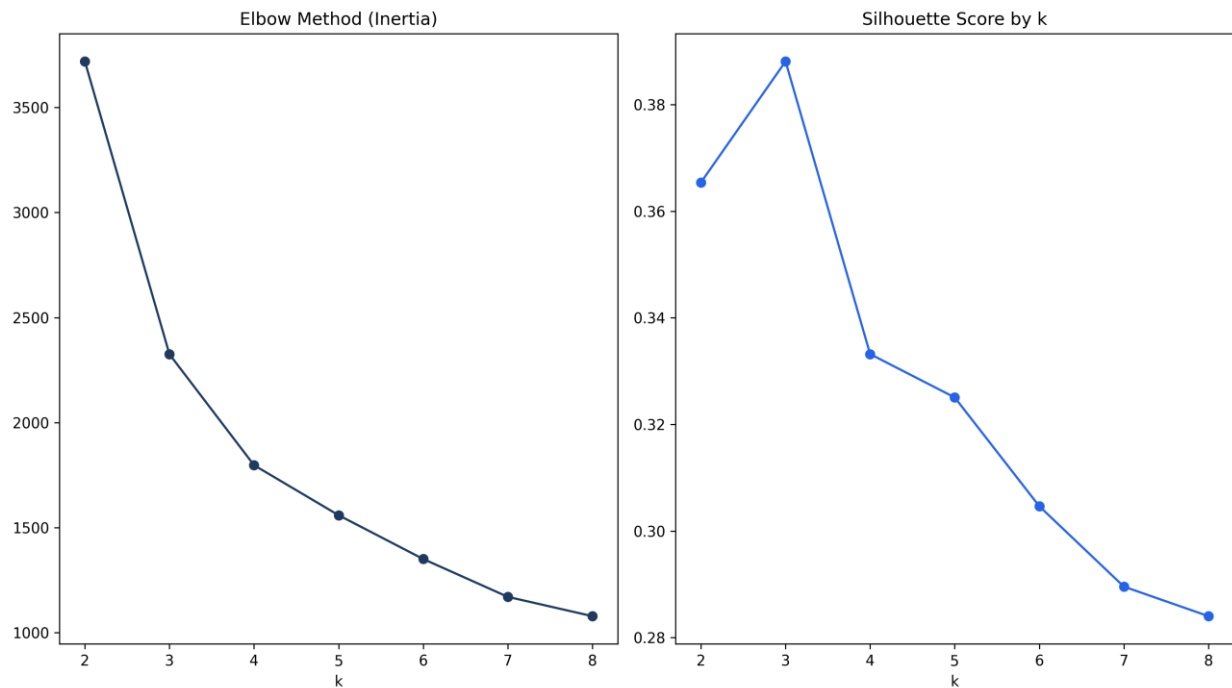


Figure 3: Two tests used to decide how many customer groups to create. The left chart shows how much "spread" is left unexplained as more groups are added (it always goes down -the question is where it stops improving much). The right chart, the Silhouette Score, measures how cleanly separated the groups are; higher is better, with anything above roughly 0.25 considered a reasonable grouping.

The cleanest possible split, statistically, is 3 groups. However, we chose 4 groups for the final result. The 4-group option still scores comfortably above the acceptable threshold (about 0.33), and it has a practical advantage: it separates a distinct, low-purchase, long-time-since-last-visit group that is genuinely useful to flag on its own for follow-up- commercially, this "at-risk" group is worth knowing about even if it costs a small amount of statistical neatness.

4.1 The four customer groups

Table 3: Cluster Summary -Champions, Repeat Customers, Occasional, At Risk

Group	Customers	Avg. days since last purchase	Avg. purchases	Avg. total spend
Champions	376	15.6	31.6	LKR 14,072
Repeat Customers	896	18.4	27.1	LKR 9,258
Occasional	614	28.5	19.6	LKR 6,356
At Risk	114	209.1	15.9	LKR 5,481

- Champions (376 customers) - buy often, spend the most, and buy recently. These are the customers to protect and reward.
- Repeat Customers (896 customers) - the largest group. Solid, regular buyers who are not yet top spenders but are consistently active.
- Occasional (614 customers) - buy less often and spend less, but are still reasonably recent.
- At Risk (114 customers) - the standout group: on average, it has been over 200 days since their last purchase, more than ten times longer than any other group. This is the group most likely to be lost for good without action.

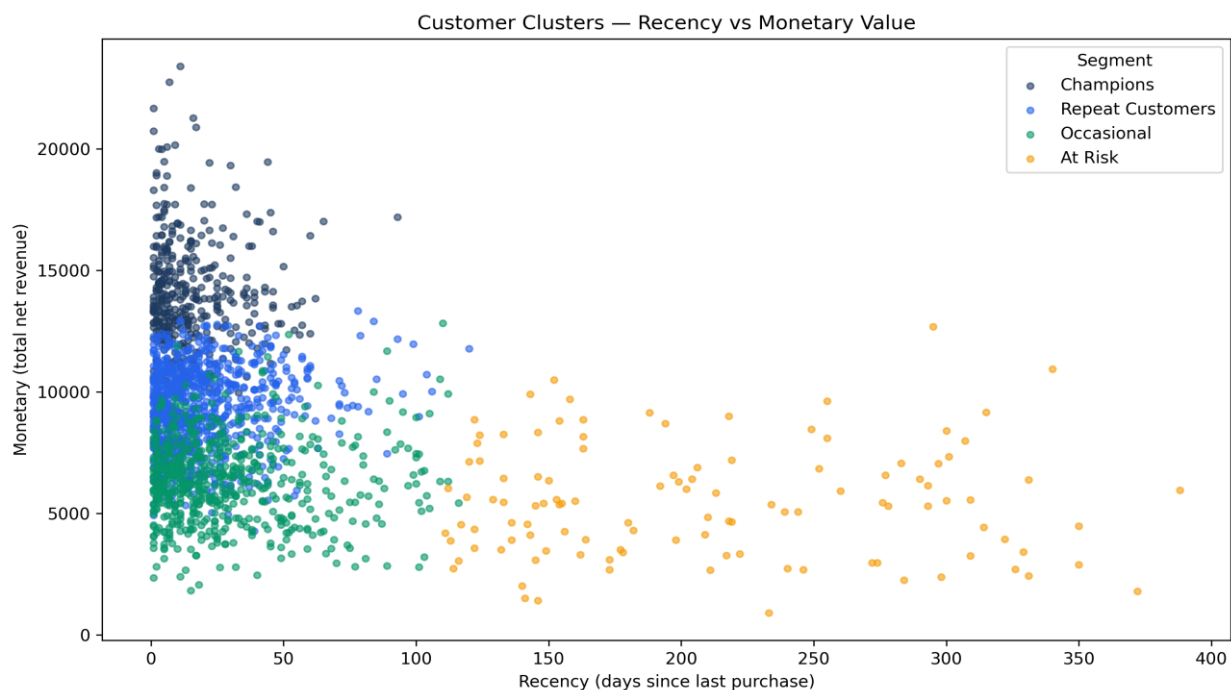


Figure 4: Every customer plotted by days since last purchase and total spent. The amber "At Risk" points sit far to the right, showing how clearly separated this group is from the rest.

5. Estimating What Each Customer Is Worth Long-Term

Beyond looking at past spending, businesses often want to estimate a customer's future value -known as Customer Lifetime Value, or CLV. This is not a measured fact; it is a forward-looking estimate built from a simple formula:

$$CLV = \text{Average Order Value} \times \text{Purchases per Year} \times \text{Assumed Customer Lifespan}$$

The one assumption in this formula is the "lifespan" -how many years we expect a typical customer relationship to last. Our transaction data only covers two years, so we cannot observe a full customer lifespan directly. We used 3 years as a simple, round planning horizon -deliberately a modest step beyond what we've actually observed, rather than an optimistic guess.

Table 4: CLV Summary Statistics

CLV statistic	Value
Median estimated lifetime value	LKR 10,994
Typical range (25th–75th percentile)	LKR 7,847 – LKR 14,768
Highest estimated lifetime value	LKR 38,649
Assumed lifespan used	3 years

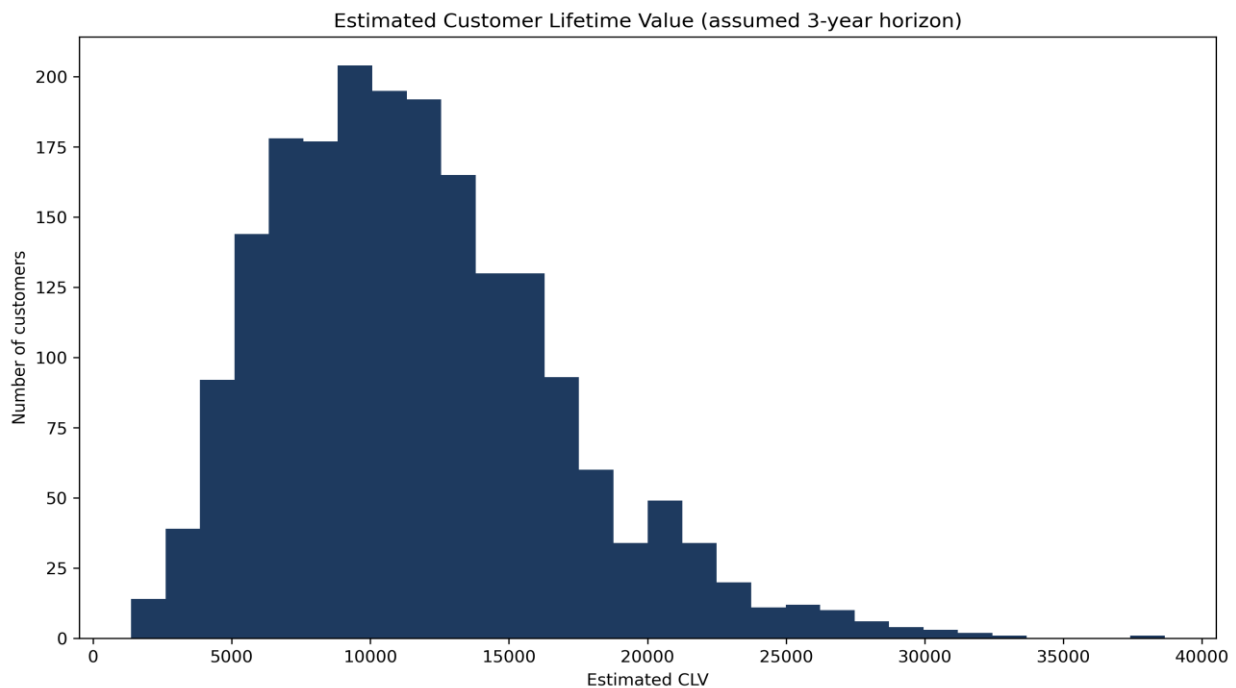


Figure 5: Estimated lifetime value across all 2,000 customers. Most customers cluster in the LKR 5,000–15,000 range, with a smaller number of high-value outliers extending the tail to the right.

Because this figure depends on the 3-year assumption, it should be used to compare customers against each other (who is relatively more valuable) rather than treated as a guaranteed future revenue number.

6. Customers Who Have Gone Quiet

Every customer in this dataset has made at least 6 purchases -so a basic "did they buy more than once?" check isn't useful here; everyone passes it. A more meaningful question is: who has stopped buying recently, even though they used to be active?

We defined a customer as "inactive" if more than 90 days have passed since their last purchase. This threshold is not arbitrary -the typical (median) gap between purchases across all customers is only about 16 days, so going quiet for 90 days -more than 5 times the normal gap -is a meaningful warning sign, not routine variation.

Table 5: Inactive Customer Summary (90+ Day Threshold)

Inactivity check	Result
Customers inactive for 90+ days	148 customers (7.4% of the base)
Historical revenue tied to these customers	LKR 901,341

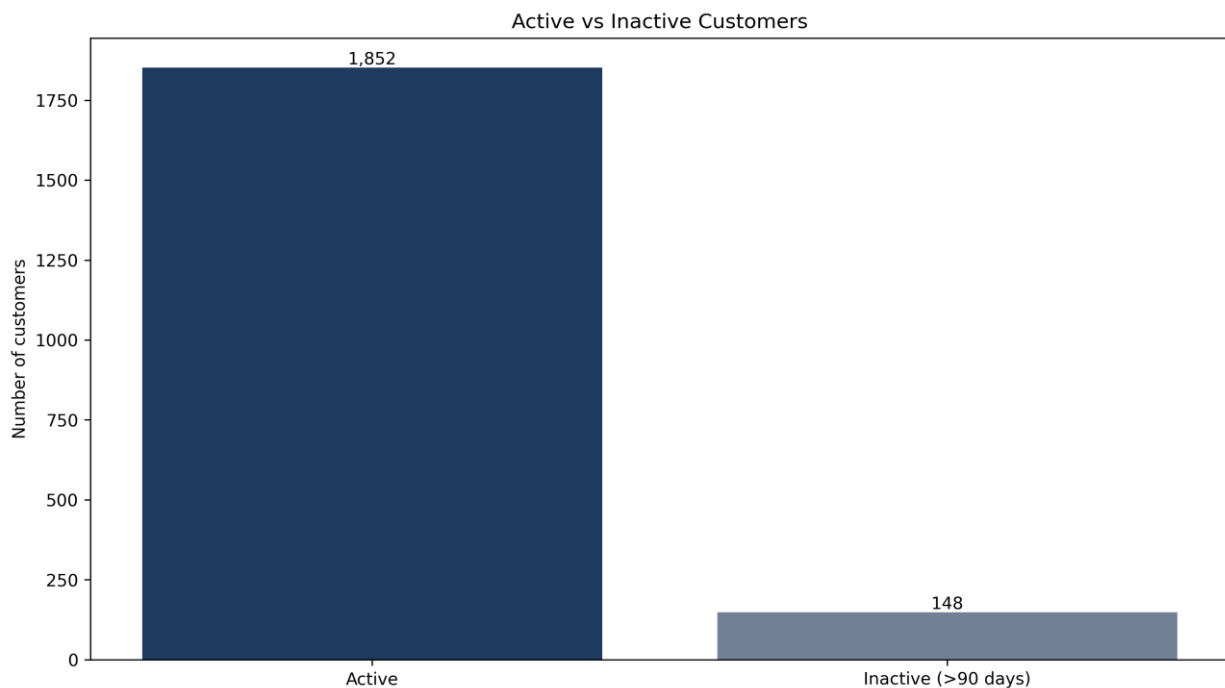


Figure 6: Active customers (bought within the last 90 days) versus inactive ones. The inactive group is a minority, but represents nearly LKR 1 million in past spending that is currently generating no revenue.

These 148 customers are not new or low-spending; collectively, they have already proven they will spend close to LKR 1 million with us. A targeted win-back campaign (e.g., a reminder offer or check-in) is likely to be far cheaper than acquiring 148 brand-new customers of similar value.

7. How Dependent Are We on Our Biggest Customers?

It's important to know whether the business relies heavily on a small handful of big spenders -because losing even one or two of them would then hurt disproportionately. We ranked every customer by total spend and checked what share of all revenue the top spenders account for.

Table 6: Revenue Concentration Among Top Customers

Revenue concentration check	Result
Share of revenue from top 10% of customers (200 people)	17.3%
Share of revenue from top 20% of customers (400 people)	31.2%

A well-known rule of thumb in business (the "80/20 rule") suggests that roughly 80% of revenue often comes from just 20% of customers. Here, the top 20% account for only 31.2% -far below that pattern. In practical terms, revenue is spread broadly across the customer base rather than concentrated in a small number of accounts. This is a healthier, lower-risk position than the classic 80/20 pattern, and it differs from the more concentrated pattern typically seen when looking at revenue by product rather than by customer.

7.1 Top 10 customers by total spent

Table 7: Top 10 Customers by Total Spend

Customer ID	Days since last purchase	Purchases	Total spent (LKR)	Value tier	Group
C01284	11	32	23,400.90	High Value	Champions
C00134	7	37	22,750.37	High Value	Champions
C01019	1	28	21,665.72	High Value	Champions
C00231	16	42	21,269.99	High Value	Champions
C01132	17	34	20,894.44	High Value	Champions
C01969	1	36	20,733.87	High Value	Champions
C01733	9	31	20,157.01	High Value	Champions
C00079	6	30	20,081.06	High Value	Champions
C01158	3	27	19,992.51	High Value	Champions
C00929	4	43	19,982.56	High Value	Champions

Reassuringly, every one of the top 10 customers by spend also falls into the High Value tier, and the Champions group identified earlier in this report -the different methods used throughout this analysis agree with one another, which adds confidence to the overall findings.

Summary of Findings

This looks at 2,000 customers and the 50,000 purchases they made to answer a simple business question: who are our best customers, who is slipping away, and how dependent are we on a small number of big spenders? Every number below comes directly from the transaction data -nothing is guessed.

- We grouped customers by how recently, how often, and how much they buy. This split the customer base into 504 High Value, 845 Medium Value, and 651 Low Value customers.
- The current business labels (Premium / Standard / Budget / New) don't line up perfectly with actual buying behavior -a meaningful number of "Budget" or "Standard" customers are, in practice, spending like top customers.
- An automated grouping method confirmed four natural customer types: Champions (376 customers), Repeat Customers (896), Occasional buyers (614), and an At-Risk group (114) who have gone quiet.
- 148 customers (7.4% of the base) have not purchased in over 90 days. Between them, they have historically generated LKR 901,341 in revenue -money that is now at risk of being lost for good.
- The top 10% of customers generate 17.3% of all revenue, and the top 20% generate 31.2%. This is a healthy, well-spread revenue base -the business is not overly reliant on a handful of big accounts.

The customer base is broad and reasonably healthy. Still, there is a clearly identifiable at-risk group worth a proactive win-back effort, and the existing Premium/Standard/Budget labels should not be used on their own to judge customer value.

Part II

Forecasting Revenue

1. What This Is About

Every transaction in the sales dataset was added up by calendar month, producing a single series of 24 monthly revenue totals. No customer- or product-level detail was used here -this module looks purely at the company's total revenue over time, and asks: based on the pattern so far, what should we expect over the next six months?

1.1 Why This Particular Forecasting Method

There are several common ways to forecast revenue -moving averages, exponential smoothing, seasonal decomposition, and more advanced models such as ARIMA. The choice here came down to one practical constraint: with only 24 months of history, the data contains exactly two full annual cycles.

- More sophisticated seasonal models typically need at least two full yearly cycles just to get started, leaving no spare data left over to check whether they actually work.
- A classical "trend plus seasonal pattern" approach needs no more data than we already have, and - importantly -every step of it can be inspected and explained, rather than being a black box.

This classical approach was therefore used as the main method, with a simple "flat average" forecast used as a baseline for comparison -a bar the real model needs to clear to justify its extra complexity.

2. Is There a Trend or a Seasonal Pattern?

Before fitting any model, we simply plotted the 24 months of revenue to see what patterns were visible.

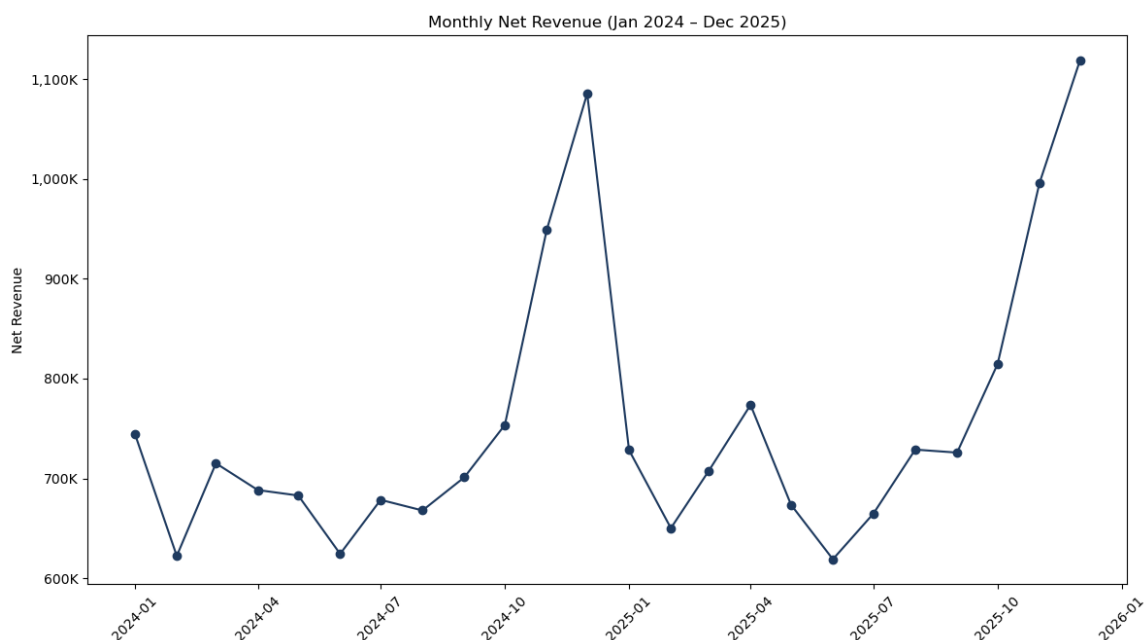


Figure 1: Monthly net revenue, January 2024 to December 2025.

The chart shows a mild overall upward trend, along with a sharp, repeating spike every November and December in both years, followed by a drop in January. We checked that this was a genuine yearly pattern -not a one-off -by averaging revenue across each calendar month, combining both years.

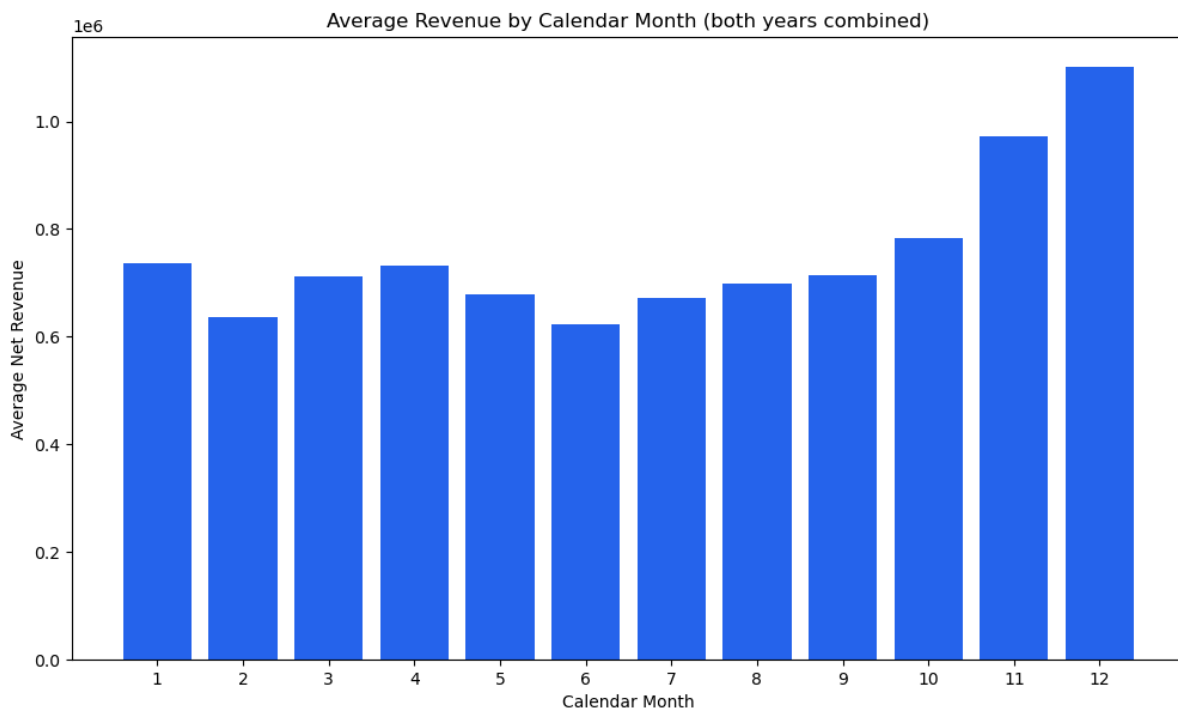


Figure 2: Average net revenue by calendar month, both years combined.

Because the November/December spike showed up consistently in both years rather than just once, we concluded that explicitly adjusting for seasonality was justified, rather than fitting a plain trend line and ignoring the yearly pattern.

3. How the Forecast Was Built

3.1 Setting Aside Test Data

To judge the forecast honestly, the last 3 months of the series (October–December 2025) were set aside and never shown to the model while it was being built. This lets us check how well the model predicts data it has genuinely never seen -the fairest possible test.

As a baseline for comparison, we also calculated a "naive" forecast: simply the flat average of the last 3 training months, carried forward unchanged. Any real forecasting model needs to clearly beat this simple benchmark to be worth using.

3.2 The Four Steps of the Model

- Step 1 -Estimate the trend: a 12-month moving average was applied to smooth out the seasonal spikes and reveal the underlying growth trend.
- Step 2 -Measure the seasonal pattern: each month's actual revenue was compared against the trend for that month, then averaged by calendar month. This produced a "seasonal index" -for example, December typically runs about 45% above trend, while June runs about 19% below trend.
- Step 3 -Remove seasonality and fit a straight trend line: the revenue series was adjusted to strip out the seasonal effect, and a simple straight-line trend was fitted to what remained, using only the training months.

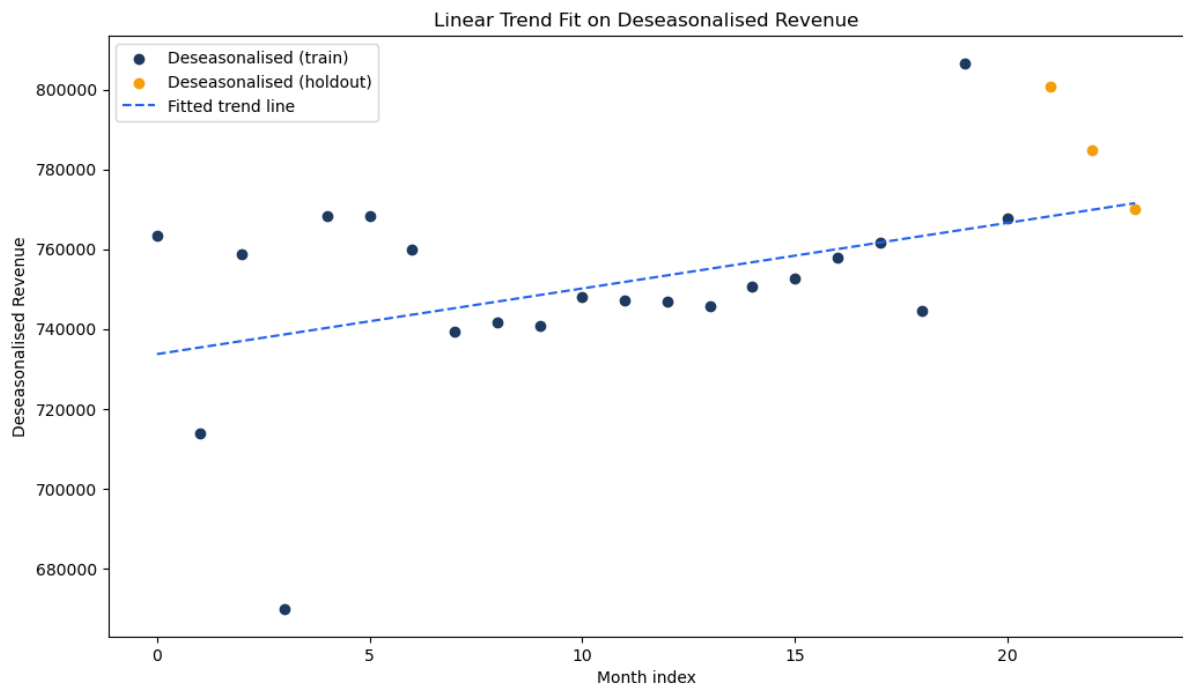


Figure 3: Straight-line trend fitted to the deseasonalised revenue series (training and holdout points shown separately).

The fitted line rises steadily through both the training and holdout points without any visible bend, which supports treating a straight-line trend as a reasonable fit for this data.

- Step 4 -Rebuild the forecast: the trend line was extended forward into the test months, then multiplied back by the relevant month's seasonal index to reintroduce the seasonal pattern.

4. How Accurate Was the Forecast?

We compared the naive baseline and our trend-plus-seasonal model against the actual revenue from the 3 held-out months (October–December 2025), using three standard accuracy measures. The most intuitive of the three is MAPE (Mean Absolute Percentage Error), which simply says, on average, how far off the forecast was, as a percentage.

Table 1: Model Accuracy Metrics (MAE, RMSE, MAPE)

Model	MAE (LKR)	RMSE (LKR)	MAPE (%)
Naive (flat average)	269,784	297,288	26.38
Trend + Seasonal Index	18,167	22,150	2.06

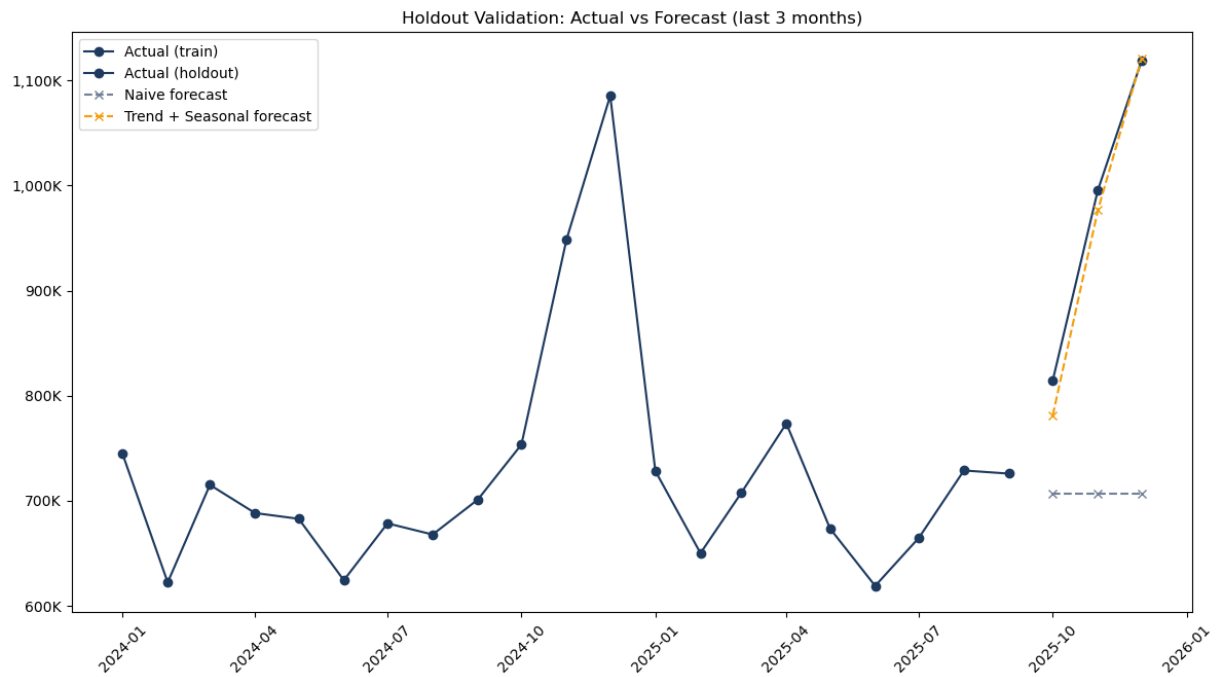


Figure 4: Actual vs. forecast revenue over the 3-month holdout period.

The trend-plus-seasonal model cut forecast error from 26.4% (naive guess) down to 2.1%, on data it had never seen during fitting. This is a strong, honestly tested result that justifies the extra effort of modelling seasonality, rather than just assuming next month looks like last month.

5. Checking the Model's Assumptions

A good accuracy score (Section 4) only tells us how big the errors were -it doesn't tell us whether those errors are just random noise, or whether they still contain a pattern the model failed to capture. To check this properly, we ran four standard diagnostic tests on the model's residuals (the difference between actual and predicted revenue), using the model refitted on the complete 24-month series.

5.1 Do the Errors Change Over Time?

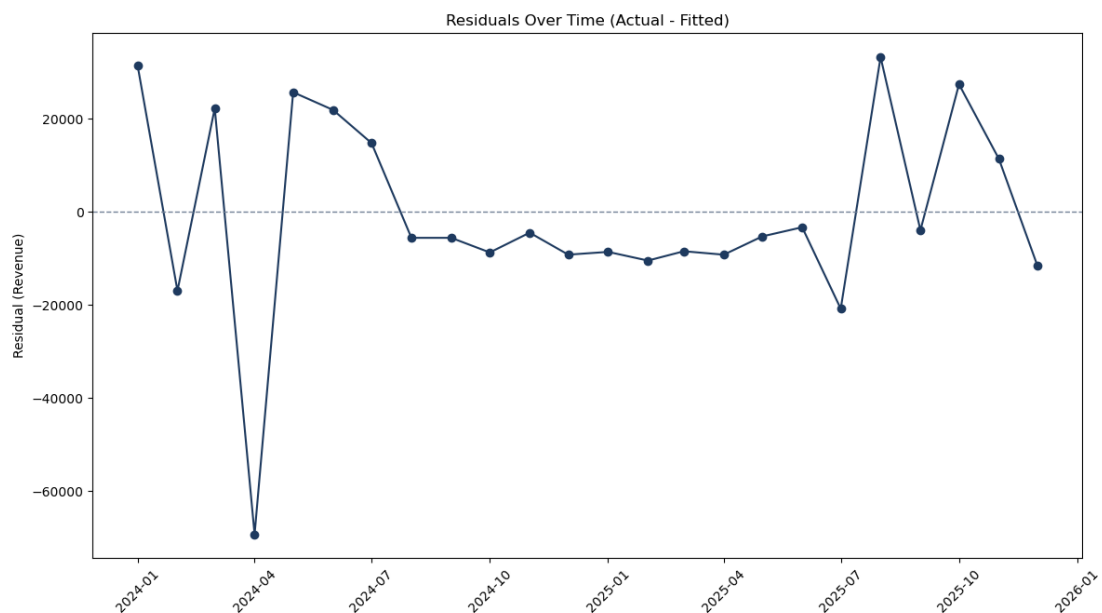


Figure 5: Residuals (actual minus predicted) plotted over time.

Ideally, these points should scatter randomly around zero with no obvious pattern. Here, the early 2024 months show noticeably larger errors than the rest of the series. This is a known side-effect of the 12-month moving average used to estimate the trend -it needs data on both sides of each point, so its earliest estimates are naturally less reliable. We state this openly as a genuine limitation of the method, rather than leaving it out.

5.2 Are Errors in Nearby Months Related? (Autocorrelation)

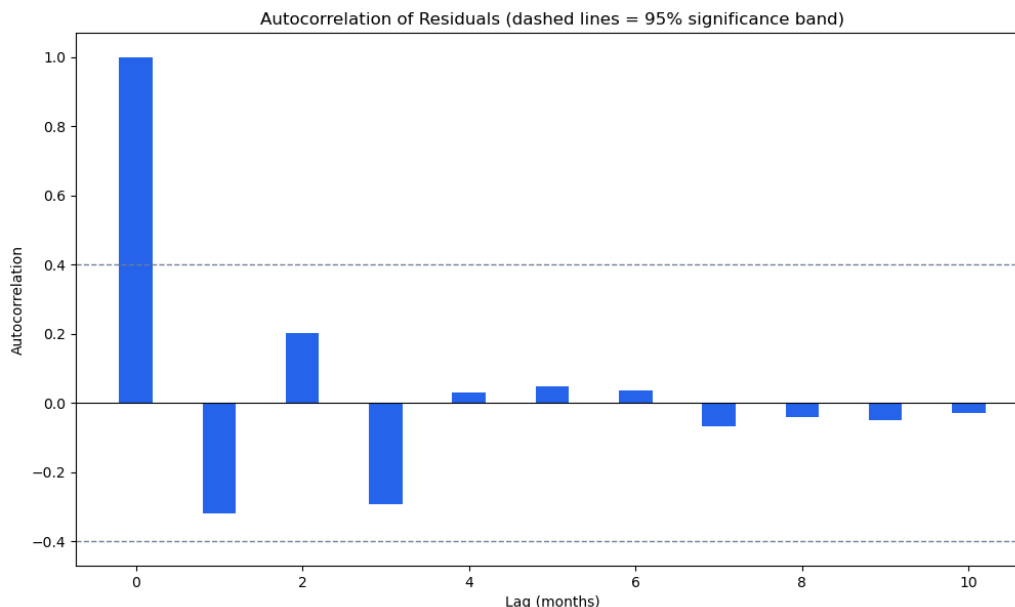


Figure 6: Autocorrelation of residuals, with a 95% significance band.

This test checks whether one month's error tends to predict the next month's error -which would suggest the model missed some leftover structure. The result (lag-1 correlation of -0.32) fell within the range we'd expect from randomness. However, with only 24 data points, this test has limited power to detect a real problem even if one exists, so we treat this as inconclusive rather than a clean pass.

5.3 Are the Errors Evenly Distributed? (Normality)

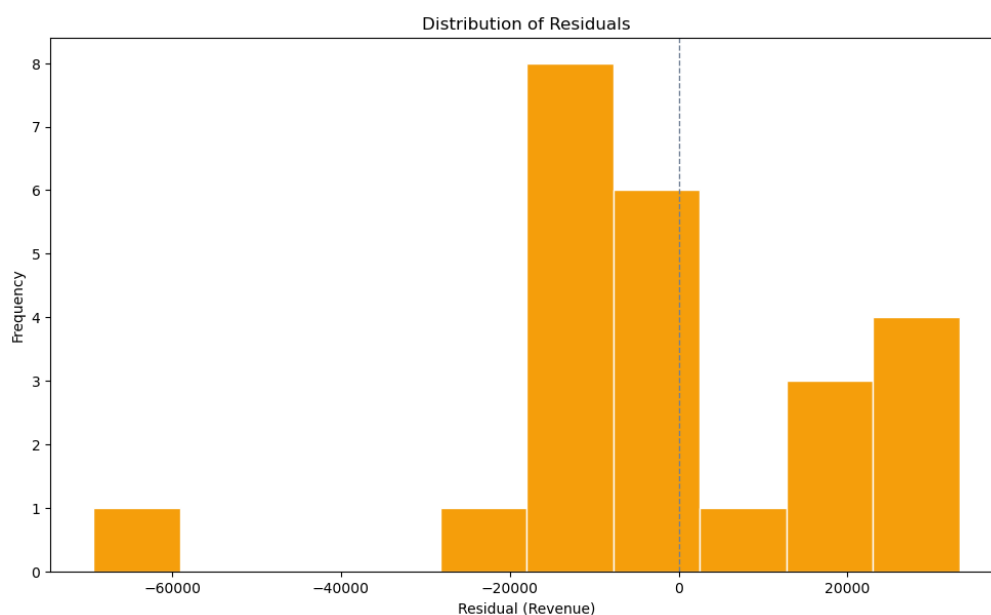


Figure 7: Distribution of residuals.

A statistical test (Shapiro-Wilk) was used to check whether the errors are roughly evenly spread around zero in the typical bell-curve shape -the assumption behind the best/worst-case range built in Section 6. The test returned a result well below the usual 0.05 cut-off ($p = 0.0036$), meaning the errors are not evenly distributed. The histogram above is not a clean, symmetric bell shape, which lines up with the larger early-2024 errors noted in Section 5.1. This is a genuine limitation: the forecast range in Section 6 should be read as an approximate guide, not a mathematically precise range.

5.4 Is the Error Size Consistent Over Time? (Homoscedasticity)

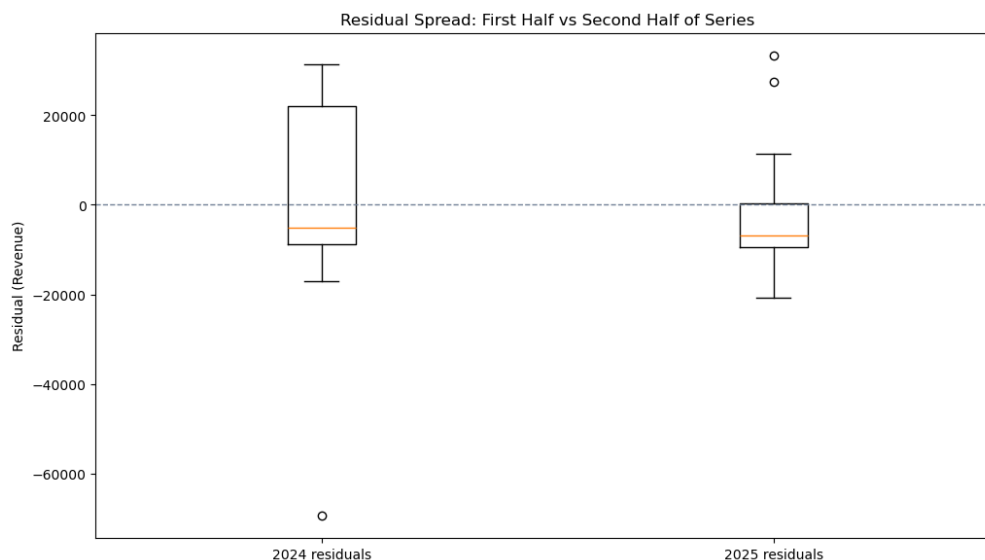


Figure 8: Residual spread, first half (2024) vs. second half (2025) of the series.

We split the errors into the first 12 months (2024) and last 12 months (2025) to check whether the typical error size stayed consistent. The 2024 half had a visibly larger spread than the 2025 half. A formal test (Levene's test) did not find this difference to be statistically significant -but with only 12 data points in each half, this result is best read as inconclusive rather than a confirmed pass.

5.5 Putting the Four Checks Together

Table 2: Summary of Residual Diagnostic Checks

Check	Result	Verdict
Errors changing over time	Larger errors in early 2024	Limitation noted
Autocorrelation (lag-1)	-0.32, within normal range	No issue found (limited power, n=24)
Normality of errors	$p = 0.0036$	Failed -errors not evenly spread
Consistent error size	$p = 0.2431$	Not significant, but uneven (n=12/group)

Taken together: two checks were inconclusive simply because there isn't much data to test with, one flagged a known weak spot in the early months, and one -normality -failed outright. None of these findings invalidate the forecast itself, but they set honest limits on how precisely the ranges in Section 6 should be interpreted.

6. The Six-Month Forecast

The trend line was refitted using the complete 24 months of data (rather than the training-only slice used for testing in Section 4) and extended six months forward, from January to June 2026. Each month's forecast was then adjusted using its seasonal index.

Rather than picking an arbitrary plus-or-minus percentage for the best/worst-case range, we used the model's own historical error size (LKR 21,537, about 2.9% of average monthly revenue). This ties the range directly to how accurate the model has actually proven itself to be. As noted in Section 5.3, this range assumes fairly evenly-spread errors -an assumption the data did not fully support -so it should be read as an approximate range rather than an exact statistical interval.

Table 3: Base, Worst-Case, and Best-Case Revenue Forecast (Jan–Jun 2026)

Month	Worst Case (LKR)	Base Case (LKR)	Best Case (LKR)
2026-01	739,228	760,765	782,302
2026-02	660,423	681,960	703,497
2026-03	717,404	738,941	760,478
2026-04	786,398	807,935	829,472
2026-05	679,057	700,594	722,131
2026-06	620,671	642,208	663,745

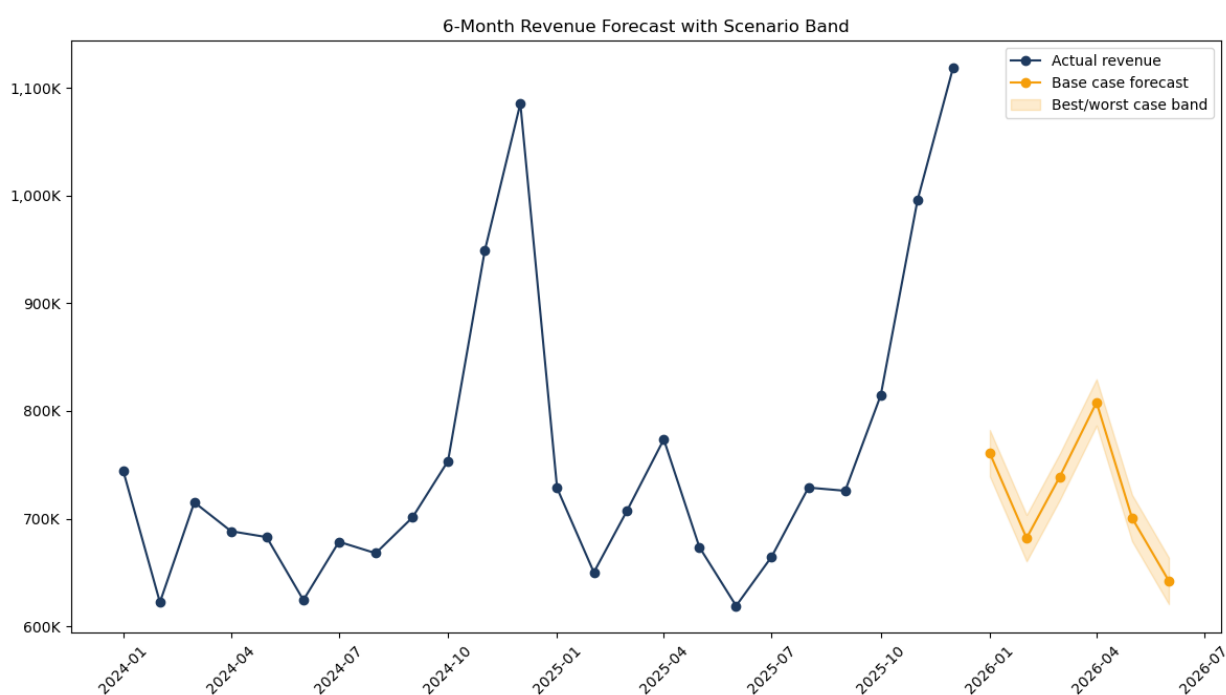


Figure 9: Historical revenue and the six-month forward forecast, with best/worst-case range.

The forecast reflects the same seasonal pattern identified in Section 2 -April is forecast higher than the surrounding months due to a mild seasonal lift, while June is forecast as the lowest month of the six, consistent with the historical mid-year dip.

Summary of Findings

This explains how we forecast the company's monthly revenue for the next six months, and -just as importantly -how much we can trust that forecast. The goal was not just to produce a number, but to test whether the model's own assumptions actually hold up, rather than presenting a forecast at face value.

We used 24 months of transaction history (January 2024 to December 2025), grouped into monthly revenue totals. Over that period, monthly revenue ranged from around LKR 619,000 to LKR 1,118,000.

- A forecasting model that accounts for both the overall growth trend and the repeating yearly pattern (e.g., a December spike) cut forecast error from 26.4% (a simple "assume next month equals last month" guess) down to just 2.1%, tested on 3 months of data the model never saw.
- Using that model, we forecast revenue for January–June 2026, with a best-case, base-case, and worst-case range for each month.
- We then stress-tested the model with four standard statistical checks. Two were inconclusive due to the limited amount of data, one showed a known weak spot in the early months, and one -the assumption that errors are evenly spread -did not pass. None of these findings break the forecast, but they set honest limits on how precisely it should be read.

The forecasting model performs far better than a naive guess and can be used with confidence for planning purposes, but the six-month forecast range should be treated as an approximate guide rather than a statistically exact prediction -a limitation we state openly rather than hide.

Overall Conclusion & Recommendations

Taken as a whole, this study gives the business two complementary views of the same underlying transaction data: who its customers are and how much they are worth (Part I), and where total revenue is headed over the next two quarters (Part II). Both analyses were built to be honest about their own limitations rather than to oversell their precision, and both point to a business that is fundamentally healthy but has specific, addressable risks.

Key Takeaways

- Customer value should be judged by behavior (RFM tier and cluster group), not by the legacy Premium/Standard/Budget/New label, which frequently disagrees with actual spend.
- A 114-customer At-Risk group and a broader 148-customer inactive group (90+ days quiet, LKR 901,341 in historical revenue) represent a concrete, quantified win-back opportunity.
- Revenue is not overly concentrated in a small number of accounts (top 20% of customers = 31.2% of revenue), which is a lower-risk position than many retail businesses face.
- The trend-plus-seasonal revenue forecast is highly accurate on held-out data (2.1% MAPE vs. 26.4% for a naive guess) and provides a usable six-month planning range for January–June 2026.
- The forecast's confidence range should be treated as approximate rather than exact, since the residual diagnostics found the errors are not perfectly evenly distributed.

Suggested Next Steps

- Launch a targeted win-back campaign for the 148 inactive customers, prioritising the At-Risk cluster identified in Part I.
- Replace or supplement the Premium/Standard/Budget/New labelling with the behaviour-based RFM tier for service prioritisation and targeted offers.
- Use the January–June 2026 base-case forecast for operational planning, while holding the worst-case figures as a contingency floor given the noted normality limitation.
- Revisit both models once more transaction history is available - a longer time series would allow a more robust seasonal model and a more reliable customer-lifespan assumption for CLV.